



AWS VPC Endpoint Lab



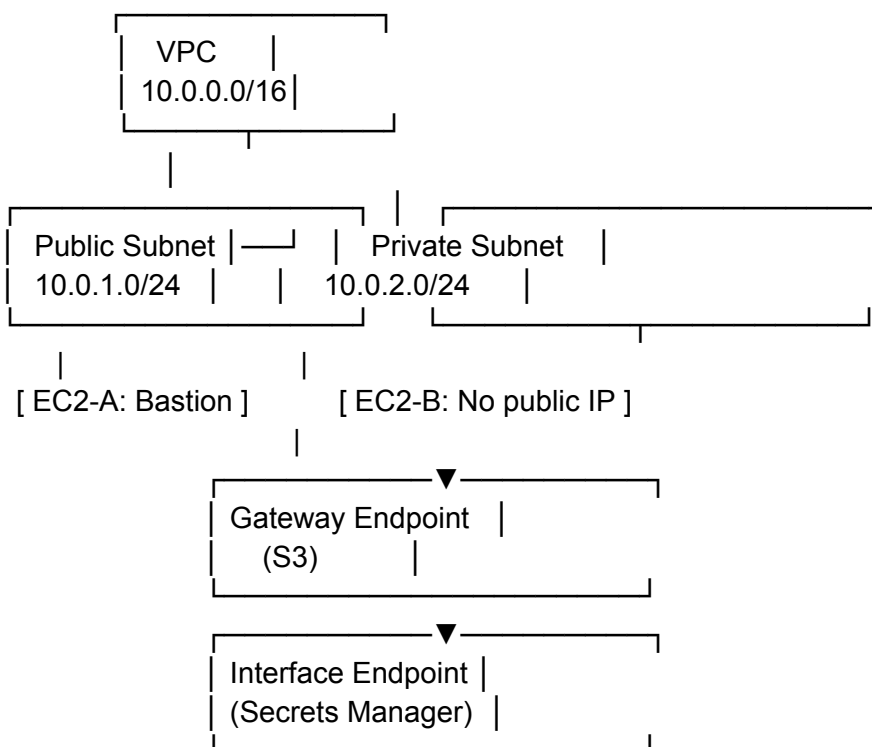
Objective

Enable **private access** to AWS services from EC2 instances in a **private subnet** using:

- **Gateway Endpoint** for Amazon S3
 - **Interface Endpoint** for AWS Secrets Manager
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Architecture Overview



Step-by-Step Instructions

Part 1: VPC & Subnets Setup

1. Create a VPC

- Name: `VPC-Endpoint-Lab`
- CIDR block: `10.0.0.0/16`

2. Create Subnets

- `PublicSubnet: 10.0.1.0/24`
- `PrivateSubnet: 10.0.2.0/24`

3. Create & Attach Internet Gateway

- Name: `IGW-Lab`
- Attach to your VPC

4. Configure Route Tables

- Public Route Table:
 - Add route: `0.0.0.0/0` → Internet Gateway
 - Associate with `PublicSubnet`
- Private Route Table:
 - No internet route yet
 - Associate with `PrivateSubnet`

Part 2: EC2 Launch & Test

1. Launch **EC2-A** (Public Subnet)

- Amazon Linux 2
- Enable public IP
- Security Group: allow SSH (port 22)

2. Launch **EC2-B** (Private Subnet)

- Amazon Linux 2
- No public IP
- Same key pair
- Security Group: allow SSH from EC2-A and allow HTTPS (port 443)

3. **SSH Access**

- SSH into EC2-A
- From EC2-A, SSH into EC2-B

Part 3: Gateway Endpoint (for S3)

Step 1: Create S3 Bucket

```
aws s3 mb s3://vpc-endpoint-lab-<your-name>
```

Step 2: Try S3 Access from EC2-B (Fails)

```
aws s3 ls
```

 Expect error due to no internet or NAT.

Step 3: Create Gateway Endpoint for S3

1. Go to: **VPC** → **Endpoints** → **Create Endpoint**
2. Name: **S3GatewayEndpoint**

3. Service: `com.amazonaws.<region>.s3`
4. Type: **Gateway**
5. VPC: `VPC-Endpoint-Lab`
6. Attach to **Private Route Table**

✓ Adds automatic route via prefix list for S3.

Step 4: Try S3 Access Again from EC2-B

```
echo "test" > test.txt
```

```
aws s3 cp test.txt s3://vpc-endpoint-lab-<your-name>/
```

✓ Should now succeed via **private connection**.

✓ Part 4: Interface Endpoint (for Secrets Manager)

Step 1: Create a Secret

1. Go to **Secrets Manager** → **Store new secret**
 - Secret name: `MyTestSecret`
 - Key: `username`, Value: `admin`

Step 2: Create Interface Endpoint

1. Go to **VPC** → **Endpoints** → **Create Endpoint**
2. Name: `SecretsManagerEndpoint`
3. Service: `com.amazonaws.<region>.secretsmanager`
4. Type: **Interface**
5. Subnet: `PrivateSubnet`

6. Enable **Private DNS**
7. Attach security group (allow HTTPS from EC2-B)

Step 3: Test Secrets Manager from EC2-B

```
aws secretsmanager get-secret-value --secret-id MyTestSecret
```

✅ Should succeed via VPC interface endpoint.

Cleanup (Optional)

- Terminate EC2 instances
 - Delete endpoints
 - Delete S3 bucket
 - Delete Secrets Manager secret
 - Delete VPC
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✅ Summary Table

Component	VPC Endpoint Type	Traffic	Notes
Amazon S3	Gateway	Route d	Uses route table + prefix
AWS Secrets Manager	Interface	ENI	Uses private IP & DNS
